



DEPARTMENT OF PHYSICS, IIT Jodhpur

Electromagnetism and Optics (Code: PHL1010)

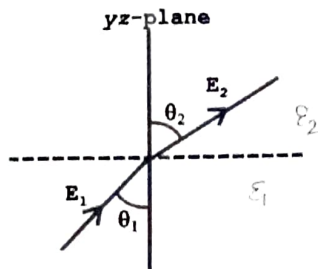
Date: 25, November 2024

Duration: 3 hr

End Sem, Semester-I

Maximum Marks: 40

1. The yz -plane forms a charge-free boundary between two media with electrical permittivities of ϵ_1 and ϵ_2 . If the electric field E_1 and E_2 in both media make angle θ_1 and θ_2 with the interface respectively, as shown in figure below, determine the relationship between θ_1 and θ_2 . (Hint: Electric displacement vector, $D = \epsilon E$)



$$D = \epsilon E$$

$$\epsilon_2 (E_2 \cos \theta_2) = \epsilon_1 (E_1 \cos \theta_1)$$

$$E_1 \sin \theta_1 = E_2 \sin \theta_2$$

$$\epsilon_1 \tan \theta_1 = \epsilon_2 \tan \theta_2$$

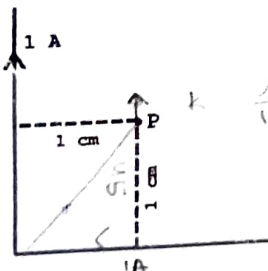
[2]

2. Two conducting spheres of radii r_1 and r_2 ($r_2 < r_1$) have charges Q_1 and Q_2 respectively. If both the spheres are connected through a metallic wire, what would be the charges on both spheres?

$$Q_1 \quad Q_2 \quad \text{Total } Q_1 + Q_2 \quad \frac{KQ_1}{r_1} \quad \frac{KQ_2}{r_2}$$

[2]

3. A long straight wire is bent into an L-shape, as shown in figure below. If a current of 1 A is flowing through the wire, determine the magnetic field at point P.



$$B = \frac{\mu_0 I}{4\pi r} (\sin \theta_1 + \sin \theta_2)$$

$$B = \frac{2 \times \mu_0 I}{4\pi r} (1 + \frac{1}{5})$$

[2]

4. A current I flows down a wire of radius a .

a) If it is uniformly distributed over the surface, what is the surface current density?

b) If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is J ?

5. Defining,

$$B = \mu H$$

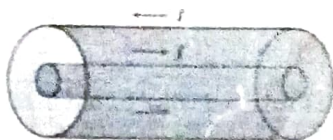
Where $\mu \equiv \mu_0(1 + \chi_m)$, and χ_m is the susceptibility of the material.

Now an infinite solenoid (n turns per unit length, current I) is filled with linear material of susceptibility χ_m . Find the magnetic field inside the solenoid.

6. Current, I flows in a wire of length L of radius " a " having a potential difference at the ends of the wire, V . Find out the energy per unit time per unit area in the wire.

7. Explain in detail the how the Ampere's law will modify in the presence of a capacitor in a circuit.

8. A long coaxial cable carries current I as shown in the figure, find the total magnetic field stored in the section L



9. A parallel beam of monochromatic light of wavelength 450 nm passes through a long slit of width 0.2 mm. Find the angular divergence in which most of the light is diffracted.
10. From the Maxwell's equations, derive the wave equations for the electric and magnetic fields in conducting medium.
11. Two Nicols prisms are so oriented that the maximum amount of light is transmitted. Determine to what fraction of its maximum value is the intensity of transmitted light reduced when the analyzer is rotated through (i) 30° (ii) 60° ?
12. The wall of a large room is covered with acoustic tile in which small holes are drilled 5.0 mm from center to center. How far can a person be from such a tile and still distinguish the individual holes, assuming ideal conditions, the pupil diameter of the observer's eye to be 4.0 mm, and the wavelength of the room light to be 550 nm?
13. How many rulings must a 4.00-cm-wide diffraction grating have to resolve the wavelengths 415.496 and 415.487 nm in the second order?
14. Red light of wavelength of 700 nm falls on a double slit separated by 400 nm. (a) At what angle is the first-order maximum in the diffraction pattern? (b) What is unreasonable about this result? (c) Which assumptions are unreasonable or inconsistent?
15. A film of soapy water ($n=1.33$) on top of a plastic cutting board has a thickness of 233 nm. What color is most strongly reflected if it is illuminated perpendicular to its surface?
16. Determine the minimum frequency which can propagate in an air-filled rectangular waveguide with the inner wall dimensions as 4 cm x 10 cm.
17. The electric field components of a plane electromagnetic wave are

$$E_x = 2E_0 \cos(\omega t - kz + \phi); E_y = E_0 \sin(\omega t - kz)$$

Draw the diagram showing the state of polarization (i.e., circular, plane, elliptical or unpolarized) when

- a). $\phi = 0$
 b). $\phi = \frac{\pi}{2}$
 c). $\phi = \frac{\pi}{4}$

18. Explain with a derivation that when an electromagnetic wave moves through a conducting medium, a complex plane wave type solution for $\vec{E}$ and $\vec{B}$ is associated with such wave equations (e.g., $\vec{K} = k + i\kappa$) and the propagation vector of the wave can be written as

$$k = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} \right]}$$

, with a frequency and conductivity dependence.

Minor Examination

Duration: 120 minutes

Marks : 40

Date: 22-02-2025 [Time: 10.30 am to 12.30 pm]

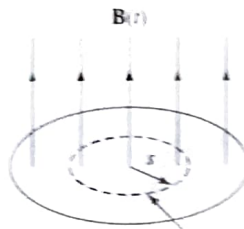
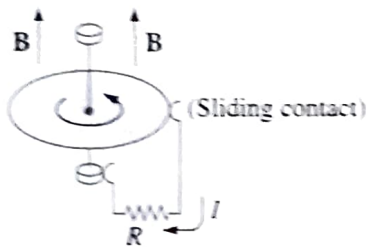
- I. Check the divergence theorem for the function $\vec{F} = 4r^3 \hat{r} + \sin \varphi \left(\frac{r}{\sin \theta} \right) \hat{\theta} + 8r \hat{\varphi}$ a solid hemi-sphere (radius "R") kept as shown in the figure. [6]



- II. (a) Find the curl of the vector $\vec{D} = r^3 \hat{r} + zr \hat{\theta} + 8rz \hat{k}$
 (b) Given $\vec{H} = 4xyz \hat{i} + (2x + 3)\hat{j} + 4y\hat{k}$, $\text{div } \vec{H}$
 (c) Evaluate $\int x^4 \delta(x - 5) dx$ [2+1+1]
- III. A spherically symmetric region having volume charge distribution in a region is given by $\rho(r) = kr$ for $R_1 \leq r \leq R_2$, where k is a constant.
 $\rho(r) = 0$ for $0 \leq r \leq R_1$ and $\rho(r) = 0$ for $R_2 \leq r \leq \infty$. Find the energy (W) associated with this charge distribution. [6]
- IV. (a) Evaluate the vector potential ($\vec{A}(r < R)$ and $\vec{A}(r > R)$), for an infinite long solenoid having radius (R), number of turns per unit length "n" and current "I" flowing through it. Find $\vec{\nabla} \times \vec{A}(r < R)$ and $\vec{\nabla} \times \vec{A}(r > R)$ using appropriate coordinate system. (Write all your assumptions)
 (b) Find the magnetic field (B) at a distance "z" from the centre of the axis of a coil having radius "R" and current "I" flows through it. [5+1]
- V. Find the electric field (E) and potential (V(r)) inside and outside a spherical shell of radius "R", having surface charge distribution $\sigma = \sigma_0 \cos \theta$. (Write down all your assumptions and boundary conditions clearly). [6]
- VI. A line-charge distribution (λ) on a circle (kept in a x-y plane), having radius "R" is given by $\lambda = \frac{q}{R}(1 + 2 \cos(\theta) + 4 \sin(2\theta))$. Find the potential due to monopole and dipole contribution. Find their electric fields also. [4+2]

VII.

- (i) Find the total bound charges for a sphere of radius R having polarization ($\vec{P} = k r \hat{r}$)
- (ii) A metal disc of radius R rotates with angular velocity ω about the vertical axis through a uniform magnetic field (B). Find the current through the resistor if one end is connected to axle and other end is connected to other edge of the disk. (iii) A uniform magnetic field ($B(t)$), pointing vertically-up fills the shaded region. If B , is changing with time, **what is the induced E field and its direction?** [2+2+2]



Indian Institute of Technology Jodhpur
Department of Physics
End-Sem Exam (PHL 1010)

Duration: 3 Hours

21/11/2025

Max. Marks: 55

Instructions:

- This paper carries seven questions. Attempt all questions. You may like to use formula given at the end of the paper.
- Keep solution of all parts together for the same question. Start every question on a fresh paper.

1. (a) Is $\vec{E} = (r \cos^2 \theta) \hat{r} - (r \cos \theta \sin \theta) \hat{\theta} + 3r \hat{\phi}$ a possible electric field?
 (b) The electric field inside a sphere of linear dielectric in an otherwise uniform electric field $\vec{E}_0$ is $\vec{E} = \frac{1}{1 + \chi_e/3} \vec{E}_0$. Write down the magnetic field $\vec{B}$ inside a sphere of a linear magnetic material in an otherwise uniform magnetic field $\vec{B}_0$.
 (c) Describe the state of polarization when the electric fields of two waves are given by $\vec{E} = \hat{x} E_0 \cos(kz - \omega t) - \hat{y} E_0 \sin(kz - \omega t + \pi/4)$.
 (d) Electric field associated with an EM wave is given by $\vec{E} = \vec{E}_0 \exp i(\omega t - kx/2 - kz\sqrt{3}/2)$. Determine the direction of propagation of the wave and the nature of the associated wave front, giving correct figure.
 (e) Plane $z = 0$ defines interface between linear medium 1 ($z < 0$) with $\mu_r = 6$ & linear medium 2 ($z > 0$) with $\mu_r = 4$. There exist a surface current $\vec{K}_s (= \hat{y} / \mu_0 \text{ mA/m})$ at $z = 0$. If $\vec{B}_2 = 5\hat{x} + 8\hat{z} \text{ mWb/m}^2$, find $\vec{H}_2$. Now calculate $\vec{H}_1$ & $\vec{B}_1$ using appropriate boundary conditions. [1+1+2+3+3]
2. Consider infinitely long rectangular metal pipe (sides a and b). Cross section of the pipe is shown in Figure 1. Metal plates at $x = 0$, $x = b$ and $y = 0$ are grounded, whereas metal plate at $y = a$ is maintained at a constant potential V_0 . Find the potential inside the rectangular region. [5]
3. A long cylindrical metal tube of radius b , and a coaxial metal solid cylinder of radius a ($a \ll b$) stand vertically in a tank of dielectric oil (susceptibility χ_e , mass density ρ). The inner cylinder is maintained at potential V , and the outer tube is grounded (Fig. 2). To what height (h) does the oil rise, in the space between the metal tube and the solid cylinder? [6]
4. (a) A conducting loop of mass m , radius a and resistance R is suspended at the end of a massless (non-conducting) string of length l . This pendulum swings through an inhomogeneous magnetic field $\vec{B}$, which is perpendicular and out of the plane of the loop. Write down the equation of motion for the pendulum.
 (b) A steady current I flows through the semi-circular loop of radius r . Point P is located at the top of the vertical diameter of the semi-circular loop (Fig. 3). Find the magnetic field at point P. [4+4]

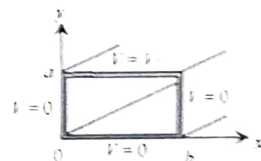


Fig. 1

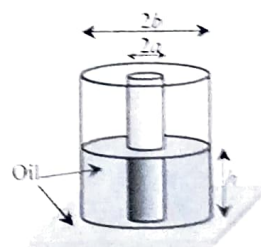


Fig. 2

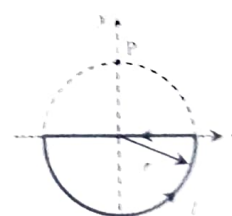


Fig. 3

5. (a) A circular loop of radius R (kept in xy -plane with its center coinciding at the origin) carries a steady current I . Find the magnetic field at a distance z above the center of the loop. Use this result for the following problem: A hollow sphere (radius R , center at the origin) carrying uniform surface charge density σ is rotating at an angular velocity $\vec{\omega}$ about the z -axis (see Fig. 4.a). Find the magnetic field at P (north pole).

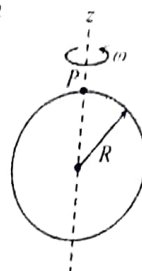


Fig. 4(a)

- (b) A long solenoid of radius $2R$ contains another long coaxial solenoid of radius R . Their coils have the same number of turns per unit length and initially both carry no current. At the same instant, currents start increasing linearly with time in both the solenoids. At any moment, current flowing in the inner coil is twice as large as the current flowing in the outer coil, whereas their directions are the same. As a result of increasing currents, a charged particle (mass m , charge q), initially at rest between the solenoids, starts moving along a circular trajectory (Fig. 4b). What is the radius r of the circular trajectory? [5+3]

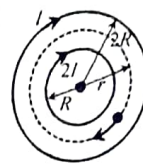


Fig. 4 (b)

6. (a) A cylinder of radius R and infinite length is made of permanently polarized dielectric. The polarization is given by $\vec{P} = ar^2$, where a is a positive constant. The cylinder rotates around its axis with an angular velocity $\vec{\omega}$. (i) Find the electric field inside and outside the cylinder. (ii) Find the magnetic field inside and outside the cylinder. (iii) Find the total electromagnetic energy stored per unit length of the cylinder before the it starts spinning and while it is spinning.

- (b) Consider a uniformly magnetized ($\vec{M}$) sphere kept in a uniform external magnetic field $\vec{B}_0 = \mu_0 \vec{H}_0$. Find $\vec{B}$ and $\vec{H}$ inside the sphere. Now suppose that the sphere is not a permanently magnetized object, but made of paramagnetic material of permeability μ . Find magnetization $\vec{M}$ of the sphere when it is exposed to the same external magnetic field $\vec{B}_0$. (you may like to use parallels between electrostatics and magneto-statics) [5+3]

7. (a) A Wollaston prism made of calcite is shown in Fig. 5a. If an unpolarized light falls on this prism at a normal incidence, draw the ray propagation through the prism depicting the field vibrations. Determine the angle of deviation between the two emergent beams if the prism angle is 20° and $n_o = 1.658$, $n_e = 1.486$.

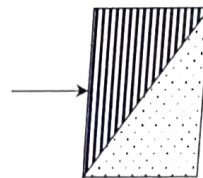


Fig. 5 (a)

- (b) An unpolarized beam falls at a linear polarizer (LP) whose pass-axis makes 45° from the horizontal. Light emerging from LP falls on a Quarter Wave Plate (QWP) made of Calcite whose optical axis is vertical (Fig. 5b). Write down the electric field after the LP, after the QWP and hence, determine the output state-of-polarization. [5 + 5]

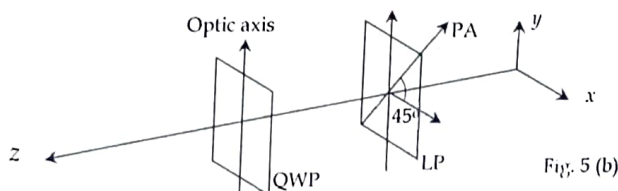


Fig. 5 (b)

$$\int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy = \frac{a}{2} \delta_{nn'}$$